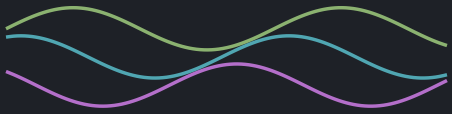
UNIVERSITY OF LAGOS

CBG 430

POPULATION GENETICS

Hardy-Weinberg and Allele Frequencies



Study Book

Lecture by Dr A.M. Oloyede

For Olufemi Racheal

Dark edition

Tap any line in Contents to jump to it

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How to use this book

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Colour key

> **Grey bar**: your lecture, word for word.

Definition

From the lecture.

Formula

From the lecture.

Note on the lecture

The lecture is unclear or disagrees with itself. Its version is kept.

Key points (added)

Short summary.

Common mistakes (added)

Errors to avoid.

Exam tip (added)

How to answer.

Memory tricks (ours)

Memory tricks, mostly from the Bible. Not from the lecture.

Page tags. [p. 4] means "from page 4 of the lecture PDF". Long lecture passages are cut with "...".

Short on time? Read Section 9, then the memory tricks, then the worked examples.

1 Gene Pool and Frequencies

1.1 Gene pool

[p. 1]

> This refers to the sum total of genes in the reproductive gametes of a population. The gene pool therefore consist of all genes in a particular population or the complete set of genetic information contained within the individuals in a population. It can be taken as a gametic pool from which samples are drawn at random to form the zygotes of the next generation. Thus the genetic relationship between an entire generation and the subsequent gen-

eration is very similar to the genetic relationship between a parent and its offspring.

Gene pool

All the genes in the gametes of a population. Zygotes of the next generation are drawn from it at random.

1.2 Gene and genotypic frequencies

[p. 1]

Gene or allele frequencies are the proportions of the different alleles of a gene being considered in a population.
Genotypic frequencies are the proportions of the different genotypes

> in the population resulting from the various combinations of these alleles.

Two kinds of frequency

Allele frequency: proportion of each *allele* in the population.
Genotypic frequency: proportion of each *genotype* in the population.

Memory tricks (ours)

Gene pool = the Pool of Bethesda. Everyone comes to the pool, and the next generation is drawn from what is in it.
Allele or genotype? Alleles are the **letters** (M, N). Genotypes are the **words** made from two letters (MM, MN, NN).

2 Calculating Allele Frequencies

2.1 Counting genes

[p. 1–2]

> If we consider an autosomal locus in a diploid, sexually reproducing species: Allelic frequencies can be measured in either of two ways. The first is simply by counting genes:
>
> $$\text{Freq of a allele} = \frac{\text{Number of a allele}}{\text{Total number of alleles}}$$
>
> The expression "frequency of" can be shortened to f( ). For example, the frequency of the a allele is written as f(a). Since the homozygotes have two of a given allele and het-

erozygotes have only one, and since the total number of alleles is twice the number of individuals (each individual carries two alleles), we can calculate allelic frequencies in the following manner.

Allele frequency

$$\text{Freq of an allele} = \frac{\text{Number of that allele}}{\text{Total number of alleles}}$$

Total alleles $= 2\times$ number of individuals.

2.2 Worked example: MN blood types

[p. 2–3]

Consider, for example, the phenotypic distribution of MN blood types (controlled by the codominant M and N alleles) among two hundred persons chosen randomly in Columbus, Ohio:
type M (MM genotype) $= 114$
type MN (MN genotype) $= 76$
type N (NN genotype) $= 10$
Total $= 200$

Step 1: count the alleles

$$p = f(M) = \frac{2(114) + 76}{2(200)} = \frac{304}{400} = 0.76$$

$$q = f(N) = \frac{2(10) + 76}{2(200)} = \frac{96}{400} = 0.24$$

Alternatively, because the frequencies of the two alleles, M and N, must add up to unity ($p + q = 1$, $q = 1 - p$, and $p = 1 - q$), if we know that $p = 0.76$, then $q = 1 - 0.76 = 0.24$.

Step 2: genotypic frequencies

$$f(MM) = \frac{114}{200} = 0.57$$

$$f(MN) = \frac{76}{200} = 0.38$$

$$f(NN) = \frac{10}{200} = 0.05$$

2.3 From genotypic frequencies to allele frequencies

[p. 2–3]

Thus, allelic frequencies can be calculated as the frequency of homozygotes, plus half the frequency of heterozygotes, as follows:

$$p = f(M) = f(MM) + \tfrac{1}{2}f(MN)$$
$$= 0.57 + \tfrac{1}{2}(0.38) = 0.76$$

$$q = f(N) = f(NN) + \tfrac{1}{2}f(MN)$$
$$= 0.05 + \tfrac{1}{2}(0.38) = 0.24$$

or $q = 1 - p = 1 - 0.76 = 0.24$.

Note that these two methods (counting alleles and using genotypic frequencies) are algebraically identical and thus give identical results.

Key takeaways (added)

- Count alleles: $(2\times$ homozygotes$)+$ heterozygotes, over $2\times$ total.

- Or: homozygote frequency $+\frac{1}{2}$ heterozygote frequency.
- Both give the same answer. $p + q = 1$.

Common mistakes (added)

- Dividing by 200 instead of 400. Each person has **two** alleles.
- Forgetting that heterozygotes give only **one** copy of each allele.
- Using genotype frequency as if it were allele frequency.

Memory tricks (ours)

Two by two, like Noah's Ark. Each person carries a pair. A homozygote brings **both** animals (2). A heterozygote brings **one** of each (so **half**).

$p + q = 1$ **is the lost sheep.** A shepherd has one whole flock. If 99 are safe ($p = 0.99$), the lost one is $q = 0.01$.

2.4 Worked example: haemoglobin, three alleles

[p. 3]

> **Hemoglobin Genotypes**

Genotype	No.	Genotype	No.
AA	2,017	SC	14
AS	783	CC	11
SS	4	**Total**	**3,002**
AC	173		

Calculation of Genotype Frequencies

$$\text{Genotype frequency} = \frac{\text{Number of individuals with the genotype}}{\text{Total number of individuals}}$$

Genotype frequencies

$$f(SS) = \frac{4}{3{,}002} = 0.0013$$

$$f(AS) = \frac{783}{3{,}002} = 0.261$$

$$f(AA) = \frac{2{,}017}{3{,}002} = 0.672$$

$$f(SC) = \frac{14}{3{,}002} = 0.0047$$

$$f(AC) = \frac{173}{3{,}002} = 0.058$$

$$f(CC) = \frac{11}{3{,}002} = 0.0037$$

Calculation of Allele Frequencies from the Number of Individuals with a Particular Genotype

$$\text{Allele frequency} = \frac{\text{Number of copies of a given allele in the population}}{\text{Sum of all alleles in the population}}$$

Allele frequencies by counting

$$f(S) = \frac{(2 \times 4) + 783 + 14}{2 \times 3{,}002} = \frac{805}{6{,}004} = 0.134$$

$$f(A) = \frac{(2 \times 2{,}017) + 783 + 173}{2 \times 3{,}002} = \frac{4{,}990}{6{,}004} = 0.831$$

$$f(C) = \frac{(2 \times 11) + 173 + 14}{2 \times 3{,}002}$$
$$= \frac{209}{6{,}004} = 0.035$$

Same result from genotype frequencies

$$f(S) = f(SS)$$
$$+\tfrac{1}{2}f(AS) + \tfrac{1}{2}f(SC)$$
$$= 0.0013 + (\tfrac{1}{2} \times 0.261)$$
$$+(\tfrac{1}{2} \times 0.0047) = 0.134$$

$$f(A) = f(AA)$$
$$+\tfrac{1}{2}f(AS) + \tfrac{1}{2}f(AC)$$
$$= 0.672 + (\tfrac{1}{2} \times 0.261)$$
$$+(\tfrac{1}{2} \times 0.058) = 0.831$$

$$f(C) = f(CC)$$
$$+\tfrac{1}{2}f(SC) + \tfrac{1}{2}f(AC)$$
$$= 0.0037 + (\tfrac{1}{2} \times 0.0047)$$
$$+(\tfrac{1}{2} \times 0.058) = 0.035$$

Exam tip (added)

With three alleles, an allele's frequency is its homozygote plus **half of every heterozygote that contains it**. Check that the three frequencies add up to 1: $0.134 + 0.831 + 0.035 = 1.000$.

3 The Hardy-Weinberg Law

3.1 Where it came from

[p. 4]

> Before the Hardy-Weinberg principle biology had operated with Mendel's genetic ratios of 3:1, 1:1, and 9:3:3:1 revealing dominant and recessive alleles. When observations were made of Mendelian traits in various populations, some dominant alleles and the corresponding phenotypes were not increasing to those predicted proportions. Brachydactily for example, is a dominant trait in humans characterized mostly by abnormally short thick fingers. In spite of

it being caused by dominant allele, the trait remained in a very low frequency in the population. Some scientist therefore suggested that dominant and recessive alleles were not segregating in a Mendelian way.
In 1908 G.H Hardy and W. Weinberg, the paradox was resolved. They independently demonstrated that a conservation of gene pool takes place from one generation to another. They demonstrated the constancy of gene and genotype frequencies.

3.2 The rule

[p. 4]

> G. H. Hardy, a British mathematician, and W. Weinberg, a German physician, independently discovered a rule that relates allelic and genotypic frequencies in a population of diploid, sexually reproducing individuals if that population has random mating, large size, no mutation or migration, and no selection. The rule has three aspects:
>
> 1. The allelic frequencies at an autosomal locus in a population will not change from one generation to the next (allelic-frequency equilibrium).
> 2. The genotypic frequencies of the population are determined in a

predictable way by the allelic frequencies (genotypic-frequency equilibrium).

3. The equilibrium is neutral. That is, if it is perturbed, it will be reestablished within one generation of random mating at the new allelic frequencies (if all the other requirements are maintained).

Hardy-Weinberg law: three aspects

1. Allele frequencies stay the same from generation to generation.
2. Allele frequencies fix the genotype frequencies.
3. If disturbed, equilibrium returns in **one generation** of random mating, at the **new** allele frequencies.

Memory tricks (ours)

Hardy = British mathematician. Weinberg = German physician. Think "**Hardy-Brit-Math**, **Wein-German-Doctor**".

Three aspects = S-P-R, "spring back". **S**ame alleles each generation, **P**redictable genotypes, **R**eset in one generation.

4 Assumptions of Hardy-Weinberg

The five assumptions

1. Random mating
2. Large population size
3. No mutation
4. No migration
5. No natural selection

4.1 Random mating

[p. 4]

> The first assumption is random mating, which means that the probability that two genotypes will mate is the product of the frequencies (or probabilities) of the genotypes in the population. If the MM genotype

makes up 90% of a population, then any individual has a 90% chance (probability $= 0.9$) of mating with a person with an MM genotype. The probability of an MM by MM mating is $(0.9)(0.9)$, or 0.81.

Deviations from random mating come about for two reasons: choice or circumstance. If members of a population choose individuals of a particular phenotype as mates more or less often than at random, the population is engaged in assortative mating. If individuals with similar phenotypes are mating more often than at random, positive assortative mating is in force; if matings occur between individuals with dissimilar phenotypes more often than at random, negative assortative mating, or

disassortative mating, is at work. Deviations from random mating also arise when mating individuals are either more closely related genetically or more distantly related than individuals chosen at random from the population. Inbreeding is the mating of related individuals, and outbreeding is the mating of genetically unrelated individuals. Inbreeding is a consequence of pedigree relatedness (e.g., cousins) and small population size.

One of the first counterintuitive observations of population genetics is that deviations from random mating alter genotypic frequencies but not allelic frequencies.
. . . Assortative mating and inbreeding will change the zygotic (geno-

> typic) combinations from one generation to the next, but will not change which alleles are passed into the next generation. Thus genotypic, but not allelic, frequencies change under nonrandom mating.

Term	Meaning
Positive assortative	Similar phenotypes mate more than at random
Negative (disassortative)	Dissimilar phenotypes mate more than at random
Inbreeding	Mating of related individuals
Outbreeding	Mating of unrelated individuals

4.2 Large population size

[p. 5]

> . . . the second assumption of the Hardy-Weinberg equilibrium is that the population is infinitely large. A large population produces a large sample of successful gametes. The larger the sample, the greater the probability that the allelic frequencies of the offspring will accurately represent the allelic frequencies in the parental population. When populations are small or when alleles are rare, changes in allelic frequencies take place due to chance alone. These changes are referred to as random genetic drift, or just genetic drift.

4.3 No mutation or migration

[p. 5]

> Allelic and genotypic frequencies may change through the loss or addition of alleles through mutation or migration (immigration or emigration) of individuals from or into a population. The third and fourth assumptions of the Hardy-Weinberg equilibrium are that neither mutation nor migration causes such allelic loss or addition in the population.

4.4 No natural selection

[p. 5]

> The final assumption necessary to the Hardy-Weinberg equilibrium is that no individual will have a reproductive advantage over another individual because of its genotype. In other words, no natural selection is occurring. (Artificial selection, as practiced by animal and plant breeders, will also perturb the Hardy-Weinberg equilibrium of captive populations.)

4.5 Why it still works in nature

[p. 5]

> In summary, the Hardy-Weinberg equilibrium holds (is exactly true) for an infinitely large, randomly mating population in which mutation, migration, and natural selection do not occur. ... Hardy-Weinberg equilibrium is approximated in natural populations for two major reasons. First, the consequences of violating some of the assumptions, such as no mutation or infinitely large population size, are small. ... Second, the Hardy-Weinberg equilibrium is extremely resilient to change because, regardless of the perturbation, the equilibrium is usually

reestablished after only one generation of random mating. The new equilibrium will be, however, at the new allelic frequencies—the Hardy-Weinberg equilibrium does not "return" to previous allelic values.

Common mistakes (added)

- Saying non-random mating changes **allele** frequencies. It changes **genotype** frequencies only.
- Saying equilibrium "goes back" to the old allele values. It settles at the **new** ones.
- Forgetting genetic drift is the small-population effect, due to chance alone.

Exam tip (added)

If asked to "state the assumptions", give all five: random mating, large population, no mutation, no migration, no natural selection.

Memory tricks (ours)

Five assumptions = five loaves. Remember five Bible women: **M**ary = **M**utation (none), **M**artha = **M**igration (none), **R**ebekah = **R**andom mating, **L**ydia = **L**arge population, **S**arah = **S**election (none).

Nonrandom mating = rearranging seats at the wedding in Cana. The seats (genotypes) change, the guests (alleles) do not.

Genetic drift = the twelve spies. A small group gave a wrong picture

of the whole land. Small samples go off by chance.

Outbreeding = Ruth. An outsider from Moab married Boaz. **In**breeding = **in** the family, like cousins.

5 Proof of the Equilibrium

5.1 Genotype frequencies from allele frequencies

[p. 5–6]

> The three properties of the Hardy-Weinberg equilibrium are that (1) allelic frequencies do not change from generation to generation, (2) allelic frequencies determine genotypic fre-

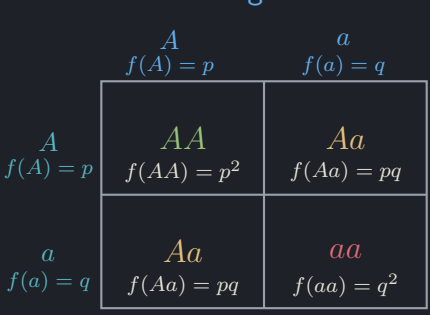
> quencies, and (3) the equilibrium is achieved in one generation of random mating. . . . In a population of individuals segregating the A and a alleles at the A locus, each individual will be one of three genotypes: AA, Aa, or aa. If $p = f(A)$ and $q = f(a)$, then we can predict the genotypic frequencies in the next generation. If all the assumptions of the Hardy-Weinberg equilibrium are met, the three genotypes should occur in the population in the same frequencies at which gametes would be randomly drawn in pairs from a gene pool. A gene pool is defined as all of the alleles available among the reproductive members of a population from which gametes can be drawn.

Hardy-Weinberg genotype frequencies

$$f(AA) = p \times p = p^2$$

$$f(Aa) = (p \times q) + (q \times p) = 2pq$$

$$f(aa) = q \times q = q^2$$



Gene pool concept of zygote formation. [p. 6]

> Gene pool concept of zygote formation. Males and females have the same frequencies of the two alleles: $f(A) = p$ and $f(a) = q$. After one

generation of random mating, the three genotypes, AA, Aa, and aa, have the frequencies of p^2, $2pq$, and q^2, respectively.

The Hardy–Weinberg law states that when a population is in equilibrium, the genotypic frequencies will be in the proportions p^2, $2pq$ and q^2.

The Hardy-Weinberg equation

$$p^2 + 2pq + q^2 = 1$$

$$p + q = 1$$

Added clarification

Why 2pq (added). The heterozygote can form in two ways: A from the father and a from the mother, or a from the father and A from the mother. Each way has chance

pq, so the total is $2pq$.

Key takeaways (added)

- $p = f(A)$, $q = f(a)$, $p + q = 1$.
- $AA = p^2$, $Aa = 2pq$, $aa = q^2$.
- $p^2 + 2pq + q^2 = 1$.

Common mistakes (added)

- Writing the heterozygote as pq. It is $2pq$.
- Mixing p^2 (homozygous dominant) with p (the allele).

Memory tricks (ours)

Adam and Eve make 2pq. Adam gives p and Eve gives q, **or** Adam gives q and Eve gives p. Two ways, so $2pq$.

Squares are the twins. p^2 and q^2 are "the same allele twice". Mixed

pair in the middle: $2pq$.
Everyone adds to one whole flock: $p^2 + 2pq + q^2 = 1$.

6 Numerical Example: MN Matings

6.1 Starting population

[p. 6]

> A numerical example of Hardy-Weinberg principle will again make use of the MN blood system. This a good characteristic in which to consider alleles in the population. The MN blood grouping is a simple Mendelian trait, the phenotype is easily tested, and the population is probably undergoing random

mating with regard to these alleles. Also, the heterozygotes can be distinguished from the homozygotes. Assume that a population has the following genotype composition at a particular locus.

0.30 MM

0.40 MN

0.30 NN

Note that in the beginning gene frequencies are 0.50 for M and 0.50 for N. Next, assume that this population undergoes random mating to produce another generation.

Check the allele frequencies

$$f(M) = 0.30 + \tfrac{1}{2}(0.40) = 0.50$$

$$f(N) = 0.30 + \tfrac{1}{2}(0.40) = 0.50$$

6.2 The mating table

[p. 6–7]

> Six different mating types relative to genotype combinations are possible, nine of them if the sexes are considered. The frequency of each mating type can be calculated because of randomness and the information known about genotype frequencies. Mendelian relationships can then be used to determine the proportion of resultant offspring genotypes in each case.

Mating	Frequency	MM	MN	NN
MM×MM	$0.30 \times 0.30 = 0.09$	0.09	—	—
MM×MN	$(0.30 \times 0.40) \times 2 = 0.24$	0.12	0.12	—
MM×NN	$(0.30 \times 0.30) \times 2 = 0.18$	—	0.18	—
MN×MN	$0.40 \times 0.40 = 0.16$	0.04	0.08	0.04
MN×NN	$(0.40 \times 0.30) \times 2 = 0.24$	—	0.12	0.12
NN×NN	$0.30 \times 0.30 = 0.09$	—	—	0.09
Offspring		**0.25**	**0.50**	**0.25**

Proportion of each genotype among the progeny [p. 7]

> NB: when genotypes of the two parents are different, the product is doubled to account for the matings being reciprocal. For example, the probability must take into account both MN females × NN males and NN females × MN males.

6.3 What the table shows

[p. 7]

> Note from the above example that the gene frequencies have not changed. They are 0.50M and 0.50N in both generations. This, exactly is the conservation of gene frequencies that is predicted under Hardy-Weinberg principle. Also note that the genotype frequencies among the progeny are now different from the parental genotype frequencies. The reason for the discrepancy is that the initial parental genotype frequencies were not in equilibrium. However only one generation of random mating is necessary to place these genotype frequencies into equilibrium.

When the gene and genotype frequencies remain unchanged from one generation to the next, the population is deemed to be in Hardy-Weinberg equilibrium.

Added clarification

Check with the formula (added). $p = q = 0.50$, so $p^2 = 0.25$, $2pq = 0.50$, $q^2 = 0.25$. These are exactly the offspring row of the table.

Key takeaways (added)

- Parents $0.30/0.40/0.30$ were **not** in equilibrium.
- After one random-mating generation: $0.25/0.50/0.25$. In equilibrium.
- Allele frequencies stayed 0.50 and 0.50.

- Different parent genotypes: **double** the product.

Common mistakes (added)

- Forgetting to double the product when the two parents differ.
- Thinking the offspring genotypes must match the parents'. Only the **alleles** are conserved.

Memory tricks (ours)

Mixed couples count twice. MN mother with NN father and NN mother with MN father are two different matings. **Different parents = double.**

0.25, 0.50, 0.25 is 1:2:1. A quarter, a half, a quarter: the same as $p^2 : 2pq : q^2$ when $p = q = 0.5$.

7 Testing the Fit

7.1 The question

[p. 7]

> There are several ways to determine whether a given population conforms to the Hardy-Weinberg equilibrium at a particular locus. However, the question usually arises when there is just a single sample from a population, representing only one generation. Can the existence of the Hardy-Weinberg equilibrium be determined with just one sample? The answer is that we can determine whether the three genotypes (AA, Aa, and aa) occur with the frequencies p^2, $2pq$, and q^2. If they do, then the population is considered to

be in Hardy-Weinberg proportions; if not, then the population is not considered to be in Hardy-Weinberg proportions.

7.2 Chi-square test, MN blood types

[p. 7]

To determine whether observed and expected allelic frequencies are the same, we can use the chi-square statistical test. In a chi-square test, we compare an observed number with an expected number. In this case, the observed values are the actual numbers of the three genotypes in the sample, and the expected values come from the prediction that the genotypes will occur in the p^2, $2pq$,

> and q^2 proportions. ... Since the critical chi-square for one degree of freedom at the 0.05 level is 3.841 (see table 4.4), we find that the Ohio population does not deviate from Hardy-Weinberg proportions at the MN locus.

	MM	MN	NN	Total
Observed numbers	114	76	10	200
Expected proportions	p^2 (0.5776)	$2pq$ (0.3648)	q^2 (0.0576)	1.0 1.0
Expected numbers	115.52	72.96	11.52	200.0
$\chi^2 = (0-E)^2/E$	0.020	0.127	0.201	0.348

Chi-square test for a sample of 200 persons, $p = 0.76$, $q = 0.24$

Note: the formula in the table

The table prints $\chi^2 = (0-E)^2/E$ with a zero. It means $(O-E)^2/E$,

where O is the observed number. Kept as written.

Note: the total

The lecture's total 0.348 is the sum of the rounded values $0.020 + 0.127 + 0.201$. Without rounding it is 0.347. Either way it is far below 3.841.

Chi-square steps

1. Find p and q from the data.
2. Expected proportions: p^2, $2pq$, q^2.
3. Expected numbers: proportion $\times$ total.
4. For each genotype: $\dfrac{(O-E)^2}{E}$.
5. Add them to get χ^2.
6. Compare with the critical value.

7.3 Degrees of freedom

[p. 7]

> ... If we calculate p from a sample, we lose one degree of freedom. However, we do not lose a degree of freedom for estimating q, since q is no longer an independent variable: $q = 1 - p$. So in the previous case, we lose two degrees of freedom—one for estimating p and one for independent categories. The general rule of thumb in using chisquare analysis to test for data fit to Hardy-Weinberg proportions is that the number of degrees of freedom must equal the number of phenotypes minus the number of alleles (in this case, $3 - 2 = 1$).

Degrees of freedom

$$\text{d.f.} = \text{phenotypes} - \text{alleles}$$

MN example: $3 - 2 = 1$.

Decision rule (from the example)

Critical χ^2 at d.f. $= 1$, 0.05 level: **3.841**.
Calculated $\chi^2 = 0.348 < 3.841$, so the population **does not deviate** from Hardy-Weinberg proportions.

7.4 Chi-square table

[p. 9]

	Probabilities						
d.f.	0.99	0.95	0.80	0.50	0.20	0.05	0.01
1	0.000	0.004	0.064	0.455	1.642	3.841	6.635
2	0.020	0.103	0.446	1.386	3.219	5.991	9.210
3	0.115	0.352	1.005	2.366	4.642	7.815	11.345
4	0.297	0.711	1.649	3.357	5.989	9.488	13.277
5	0.554	1.145	2.343	4.351	7.289	11.070	15.086
6	0.872	1.635	3.070	5.348	8.558	12.592	16.812
7	1.239	2.167	3.822	6.346	9.803	14.067	18.475
8	1.646	2.733	4.594	7.344	11.030	15.507	20.090
9	2.088	3.325	5.380	8.343	12.242	16.919	21.666

Common mistakes (added)

- Using $(O-E)$ without squaring, or dividing by O instead of E.
- Using percentages instead of **numbers** for O and E.
- Using the wrong d.f. For a normal HW test with 3 phenotypes and 2 alleles it is 1.

- Saying "accept" when χ^2 is bigger than the critical value. Bigger means **deviates**.

Exam tip (added)

Write the d.f., the critical value and a one-line conclusion: "Calculated χ^2 is less than 3.841, so the population is in Hardy-Weinberg equilibrium at this locus."

Memory tricks (ours)

Observed minus Expected, then Square, then divide by Expected. Say it like a verse: **O-E, Square, over E**.

d.f. = 3 − 2 = 1. Three phenotypes, two alleles, one degree of freedom. Count down: **3, 2, 1**.

Under 3.841 = innocent. If your

χ^2 stays below the limit, the population is not guilty of deviating. Over the limit, it deviates.

8 Dominant Traits: the PKU Example

[p. 8]

> Circumstances sometimes do not allow us to test for Hardy-Weinberg proportions. In the case of a dominant trait, for example, allelic frequencies cannot be calculated from the genotypic classes because the homozygous dominant individuals cannot be distinguished from the heterozygotes. However, we can estimate allelic frequencies by assuming that the Hardy-Weinberg equi-

librium exists and, thereby, assuming that the frequency of the recessive homozygote is q^2, from which q and then p can be estimated. If, for example, Hardy-Weinberg equilibrium is assumed for a disease such as phenylketonuria (PKU), which is expressed only in the homozygous recessive state, it is possible to calculate the proportion of the population that is heterozygous (carriers of the PKU allele).

... Thus the assumption of no selection, required for equilibrium, is violated. However, only one child in ten thousand live births has PKU. When a genotype is as rare as one in ten thousand, selection has a negligible effect on allelic frequencies. Therefore, because of the rarity of

the trait, we can assume Hardy-Weinberg equilibrium and calculate

PKU calculation

Frequency of recessive homozygote $= q^2 = \frac{1}{10{,}000} = 0.0001$

so

$$q = \sqrt{0.0001} = 0.01$$

and

$$p = 1 - q = 0.99$$

Therefore, frequency of normal homozygote:

$p^2 = (0.99)^2 \equiv 0.98$ or 98 in 100

frequency of heterozygote:

$$2pq = 2(0.01)(0.99)$$
$$\equiv 0.02 \text{ or } 2 \text{ in } 100$$

By assuming the Hardy-Weinberg equilibrium, we have discovered something not intuitively obvious: A recessive gene causing a trait as rare as one in ten thousand is carried in the heterozygous state by one individual in fifty. Obviously, the chi-square test cannot be used to verify the Hardy-Weinberg proportions since we derived the allelic frequencies by assuming Hardy-Weinberg proportions to begin with. In statistical terms, the number of phenotypes minus the number of alleles $= 2 - 2 = 0$ degrees of freedom, which precludes doing a chi-square test.

Steps for a recessive-disease question

1. Recessive frequency $= q^2$.
2. $q = \sqrt{q^2}$.
3. $p = 1 - q$.
4. Carriers $= 2pq$.
5. Normal homozygotes $= p^2$.

Common mistakes (added)

- Calling $1/10{,}000$ the value of q. It is q^2. Take the square root.
- Forgetting the 2 in $2pq$.
- Trying a chi-square test here. The d.f. is $2 - 2 = 0$, so it is not possible.

Exam tip (added)

$2pq = 2(0.01)(0.99) = 0.0198$, which the lecture rounds to 0.02, or 1 in 50. Quote both the decimal

and “1 in 50”.

Memory tricks (ours)

The 99 and the 1. Ninety-nine sheep are safe ($p = 0.99$) and one is lost ($q = 0.01$). The lost sheep is the **root**: you start from q^2 and take the **square root**.

Carriers: 2 in 100. Two hidden sheep for every hundred, which is **1 in 50**.

Recessive shows only when “two lost sheep” meet: $q \times q = q^2$.

9 Exam Cheat Sheet

Formulas

$$p + q = 1 \qquad p^2 + 2pq + q^2 = 1$$

$$f(\text{allele}) = \frac{\text{copies of the allele}}{2 \times \text{individuals}}$$

$$p = f(AA) + \tfrac{1}{2} f(Aa)$$

$$q = f(aa) + \tfrac{1}{2} f(Aa)$$

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

$$\text{d.f.} = \text{phenotypes} - \text{alleles}$$

Numbers to know

Critical χ^2, d.f. $= 1$, 0.05: **3.841**.
MN example: $p = 0.76$, $q = 0.24$.
Mating table offspring: 0.25, 0.50, 0.25.

PKU: $q^2 = 0.0001$, $q = 0.01$, $p = 0.99$, carriers ≈ 1 in 50.

Hardy-Weinberg in one breath

Infinitely large, randomly mating population, no mutation, no migration, no selection. Allele frequencies stay constant. Genotypes are p^2, $2pq$, q^2. Disturbed? Equilibrium returns in one generation at the new allele frequencies.

What changes what

- Non-random mating: changes **genotype** frequencies only.
- Small population: genetic drift, changes **allele** frequencies by chance.
- Mutation, migration, selection: change **allele** frequencies.

Which method?

Got counts of each genotype: count alleles, or $f(\text{homozygote}) + \frac{1}{2} f(\text{hetero})$.
Got only the recessive frequency: $q = \sqrt{q^2}$, then $p = 1 - q$.
Asked if a population fits HW: χ^2 test.

Memory tricks: all in one place

Noah's Ark: two by two. Homozygote full, heterozygote half.
Lost sheep: $p + q = 1$, the 99 and the 1.
Adam and Eve: two ways, so $2pq$.
Mary, Martha, Rebekah, Lydia, Sarah: Mutation, Migration, Random mating, Large population, Selection.
Twelve spies: small sample, wrong

picture: genetic drift.
O-E, Square, over E. 3, 2, 1. Under **3.841** = fits.

10 Notes on the Lecture

Wording kept as written

- "Brachydactily" is kept as written.
- $(0 - E)^2/E$ in the table is kept. It means $(O - E)^2/E$.
- The table total 0.348 is the sum of rounded values (exact: 0.347).
- "to determine whether observed and expected allelic frequencies are the same": the test actually compares **genotype** num-

bers. Kept as written.

Cut text

Where the lecture repeats itself, long passages are cut and marked with "...". Every definition, equation and example is kept in full.

Textbook tables

The lecture refers to "table 19.2" and "table 4.4" of the source textbook. The chi-square values are in Section 7.

Added clarification

Everything in boxes marked "added" (key points, mistakes, tips, memory tricks and the "Why $2pq$" note) is not from your lecturer. Use the grey-bar text when you need the lecturer's exact wording.